Gaspard BEAUDOUIN

Machine Learning Research Student

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Interests: World Models, Flow Matching, Representation Learning, Vision-Language Models, Diffusion Language Models.

Experience

AI Scientist Intern - Mistral AI Paris, France
Multimodal team *2026*

- Conduct research on Vision-Language Models and Vision Encoders.

Research Intern - Harvard University Boston, MA, USA
Harvard AI and Robotics Lab, MEE *Feb 2025 – Aug 2025*

- Conducted research under Dr. M. Wang on generative modelling and flow matching. Proposed DRFS, an inversion-free flow matching framework for text-to-image editing (*CVPR 2026*), and co-authored *SplitFlow* (*NeurIPS 2025*) for flow aggregation.
- Co-authored *PAGE-4D* (*ICLR 2026*) for pose and geometry estimation of dynamic scenes.

Machine Learning Engineer Intern - Sinequa Paris, France
Research Team *Aug 2024 – Jan 2025*

- Studied LLMs and tool-use agents through supervised fine-tuning and reinforcement learning; designed evaluations for function-calling models using BFCL, data augmentation, and templating.
- Performed distributed instruction fine-tuning with LoRA and full fine-tuning, as well as DPO, on H100 clusters for tool-use models; used Docker and Slurm workflows.

Publications

Delta Velocity Rectified Flow for Text-to-Image Editing *Accepted at CVPR 2026.*
G. Beaudouin, M. Li, J. Kim, S.-H. Yoon, M. Wang

PAGE-4D: Disentangled Pose and Geometry for 4D Perception *Accepted at ICLR 2026* .
K. Zhou, Y. Wang, G. Chen, G. Beaudouin, F. Zhan, P. Liang, M. Wang

SplitFlow: Flow Decomposition for Inversion-Free Editing Accepted at NeurIPS 2025.
S.-H. Yoon, M. Li, G. Beaudouin, C. Wen, M. Azhar, M. Wang

Education

École Normale Supérieure Paris-Saclay Paris, France
MVA Master - Mathematics, Vision, Learning *2025 – 2026*

- Coursework: Advanced Learning for Text and Graphs, Deep Learning, Optimal Transport, Probabilistic Graphical Models, Large-Scale LLM Training, Generative Modelling, Representation Learning, Stochastic Calculus, Reinforcement Learning, LLMs for Code and Proof.

École Nationale des Ponts et Chaussées - Institut Polytechnique de Paris Paris, France
Mathematics and Computer Science Department *2022 – 2026*

- Focus: Deep Learning, Machine Learning, Computer Vision, Statistics, Convex Optimization, Stochastic Processes, Advanced Programming, Control Theory, and Functional Analysis.

Fénelon Sainte-Marie Paris, France
Preparatory Classes in Mathematics and Physics *2020 – 2022*

- 1st year: PCSI; 2nd year: PSI* with focus on Physics, Mathematics, and Engineering Sciences.

Projects

Native Sparse Attention Reproduction 2026
Academic Project, Large-Scale LLM Training

- Reproduced DeepSeek-AI's Native Sparse Attention with compression, selection, and sliding-window branches; implemented a PyTorch reference transformer with RoPE, RMSNorm, SwiGLU, and GQA-style attention.
- Benchmarked long-context attention on A100-SXM4-40GB, reaching up to 8.3× forward and 3.36× backward speedup at 64k tokens; designed French Needle-in-a-Haystack experiments where NSA reached 97.6% accuracy.

Flow Matching Generalization and Memorization 2026
Academic Project, Generative Modelling

- Reproduced and extended results on closed-form Flow Matching, studying why finite-data optimal velocity fields memorize while neural networks generalize.
- Derived the Flow Matching / diffusion probability-flow ODE equivalence and validated $t_C \sim \log(n)/d$ through dimension, capacity, geometry, and Exact Flow Matching ablations.

LeJEPA Re-implementation and Analysis 2025
Academic Project

– Re-implemented LeJEPA using a Tiny ViT implemented from scratch in PyTorch (17M parameters); studied training dynamics and online linear probing behavior.

Mini GRPO 2025
Personal Project

– Reimplemented GRPO, introduced by DeepSeek-R1 and DeepSeekMath, and applied it to math reasoning tasks.

GPT Transformer from Scratch 2024
Personal Project

– Implemented and trained a PyTorch Mixture-of-Experts Transformer to generate French poetry with recent GPT-OSS techniques; generated synthetic training data and wrote notes on LLMs: From Pre-Training to RLHF (Draft).

Drug Design with Diffusion Models 2024
Academic Project, École des Ponts

– Implemented DDPM and explored DiffDock for drug binding discovery with Sanofi. Poster available.

Edge Detection in Images 2022
Academic Project, École des Ponts

– C++ research project on perceptual boundary saliency; implemented edge detection algorithms.

Skills

Languages French native, English fluent, Spanish proficient

Technical Python, PyTorch, Transformers, Docker, Slurm, C++, SQL, R, LaTeX

Academic Service

Conference Reviewing Reviewer for CVPR 2026

Miscellaneous

Sports École des Ponts football team; competitive tennis

Interests Passionate about machine learning research; actively following scientific publications